NUMPY: FUEL EFFICIENCY ANALYSIS

AIM

To calculate the average fuel efficiency of car models and determine the percentage improvement between two car models using NumPy.

ALGORITHM

1. Start.
2. Import the NumPy library.
3. Create a NumPy array containing the fuel efficiency values of car models in miles per gallon (MPG).
4. Calculate the average fuel efficiency using np.mean().
5. Select the fuel efficiency values of the two car models to compare.
6. Calculate percentage improvement using ((second model MPG - first model MPG) / first model MPG) × 100.
7. Display the average fuel efficiency and percentage improvement.
8. Stop.

PROGRAM (PYTHON)

```
import numpy as np

# Fuel efficiency of car models in miles per gallon (MPG)
fuel_efficiency = np.array([25, 30, 35, 40, 32])

# Calculate average fuel efficiency
average_efficiency = np.mean(fuel_efficiency)

# Compare Model 1 and Model 2
model1 = fuel_efficiency[0]  # 25 MPG
model2 = fuel_efficiency[1]  # 30 MPG

if model1 != 0:
    improvement = ((model2 - model1) / model1) * 100
    print("Average fuel efficiency:", round(average_efficiency, 2), "MPG")
    print("Percentage improvement:", round(improvement, 2), "%")
else:
    print("Percentage improvement cannot be calculated because Model 1 is zero MPG.")
```

INPUT

Fuel efficiency values (MPG):

Car Model	Fuel Efficiency (MPG)

Model 1	25
Model 2	30
Model 3	35
Model 4	40
Model 5	32

OUTPUT

```
Average fuel efficiency: 32.4 MPG
Percentage improvement: 20.0 %
```

RESULT

The average fuel efficiency of the five car models is 32.4 MPG. Model 2 has a 20% improvement over Model 1, increasing from 25 MPG to 30 MPG.